

David Kreuter

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Senior Infrastructure Engineer | Cloud, IaC & CI/CD Specialist

Senior Infrastructure Engineer with 10+ years of experience in designing, building, and maintaining scalable cloud platforms. Expert in infrastructure as code with Terraform, CI/CD pipeline development, and API-driven integrations across complex distributed systems. Proven ability to deliver reliable, auditable environments at scale--having architected a global provisioning framework that manages 20,000+ virtual machines with 100% compliance. Possesses a deep background in observability, security hardening, cost optimization (\$400k in direct savings), and building internal tooling that accelerates developer velocity.

WORK EXPERIENCE

M&T Bank

03/2019 - Present

Automation Engineer (formerly Systems Engineer)

- **Hybrid Cloud & GitOps:** Architected GitOps-driven CI/CD pipelines with Terraform to automate the deployment of virtual machines in Azure; implemented a pull-request-based workflow that standardized infrastructure changes through automated plan/apply cycles, ensuring consistent, auditable environments.
- **Cloud Abstraction & Redesign:** Refactored legacy on-premises automation workflows to be fully cloud-agnostic, enabling seamless deployments to Azure; designed the abstraction layer to be transparent to end users, maintaining a consistent developer interface while scaling across infrastructure providers.
- **Platform Engineering & Middleware:** Designed and developed a custom Python orchestration application as middleware between ServiceNow and the Ansible Automation Platform; implemented advanced retry logic, auto-remediation of deployment failures, and a pub/sub queue model for high-volume scaling across distributed systems.
- **Architectural Scale:** Engineered a global server-provisioning framework that deployed and managed 20,000+ virtual machines; integrated Rubrik and VMware APIs to automate backups and compute placement, ensuring 100% RTO/RPO compliance at build time.
- **Operational Efficiency:** Developed an intelligent IP provisioning engine via InfoBlox API integrations, eliminating 1,000+ annual engineer hours by creating internal tooling that streamlined workflows and resolved a multi-year infrastructure bottleneck.
- **Infrastructure Resilience:** Engineered and maintained a highly resilient infrastructure architecture spanning three identical, geo-redundant data centers to ensure maximum uptime, reliability, and disaster-recovery readiness.
- **End-to-End Hardening & Provisioning:** Developed a zero-touch build workflow that delivered fully patched and hardened Linux (Ansible) and Windows (GPO) servers; automated agent installation, drive formatting, and optional hardened SQL deployments to meet security and compliance requirements at scale.
- **Database as a Service (DBaaS):** Collaborated with Database Engineering to architect a high-performance on-premises SQL cluster, enabling a unified DBaaS offering that consolidated fragmented legacy databases and streamlined application delivery.
- **Cost Optimization:** Orchestrated the migration of 850 branch file servers within a strict 3-month window using custom PowerShell automation, avoiding \$400,000 in extended licensing fees through strategic vendor management and planning.
- **Network Security Automation:** Automated Check Point firewall rule generation and DMZ provisioning; integrated security policy updates into the server build cycle to ensure day-zero network isolation and compliance with organizational security standards.
- **Technical Leadership:** Serve as a primary mentor for the Technology Development Program (TDP), setting enterprise-wide standards for naming conventions, Git branching strategies, and unit testing for infrastructure code.
- **Risk Mitigation:** Authored Active Directory lifecycle scripts with built-in "Read-Only" dry-run modes; implemented Least Privilege permissions and robust rollback protocols for stale object cleanup.

OTG Management

07/2013 - 02/2019

Lead Network Engineer

- **Infrastructure Resilience:** Architected a highly redundant DNS solution that significantly reduced network outages and optimized global traffic performance and reliability.
- **Cloud Connectivity:** Established secure, encrypted VPN tunnels to bridge on-premises networks with cloud resources, ensuring seamless hybrid cloud operations.
- **Security Architecture:** Engineered a secure reverse-proxy solution to protect sensitive internal infrastructure, enabling online customer ordering while mitigating external threat vectors.
- **Performance Engineering:** Diagnosed and resolved a mission-critical server hang by identifying a NUMA architecture mismatch; realigned vCPU/memory topology to eliminate cross-node latency and restore system stability.

Network Engineer

- **Proactive Monitoring:** Developed service-aware alerting and observability frameworks for business-critical systems, reducing mean time to resolution (MTTR) through proactive incident detection and diagnostics.
- **Cybersecurity Governance:** Directed enterprise-wide vulnerability scanning and remediation efforts for internal and external systems to maintain a hardened security and compliance posture.
- **Network Administration:** Managed complex firewall rules and switch ACLs for a massive distributed network spanning 12 airports and 3 satellite offices.

- **Enterprise Mobility:** Successfully onboarded and managed 15,000 iPads via Meraki MDM, implementing automated security protocols including geofencing, remote wipe triggers, and custom application deployment.

Systems Administrator

- **Operational Visibility:** Modernized infrastructure monitoring and observability by designing custom Grafana dashboards to visualize real-time performance metrics, alerting, and diagnostics.
- **Disaster Recovery:** Spearheaded the deployment of Datto backup and disaster recovery appliances to ensure 100% business continuity for critical operations.

SKILLS

Languages: Python, SQL, Bash, PowerShell, Javascript, YAML, PHP

Automation/Orchestration: Terraform, CI/CD Pipelines, GitOps, Gitlab, REST API, GraphQL, Grafana, Ansible Automation Platform, Middleware Development, ServiceNow, Unit Testing

Infrastructure: Azure Cloud Services, Infrastructure as Code (IaC), DBaaS, OS Hardening (CIS/STIG), VMware vSphere, Rubrik, Nutanix AHV, Check Point Firewalls, Active Directory, NUMA Tuning

Networking: DNS, High Availability, F5 Load Balancing, VPN, BGP, Intrusion Detection, Vulnerability Scanning, VRRP, Meraki MDM

EDUCATION

Bachelor Of Science in Computer Information Systems

12/2012

SUNY Fredonia

CERTIFICATIONS

Certified Meraki Network Operator

Meraki

VOLUNTEERING & LEADERSHIP

Habitat For Humanity

Volunteer

07/2025 - Present

Tift Nature Preserve

Volunteer

01/2025 - Present

716 Paws

Thermal drone pilot for search and rescue missions.

01/2022 - Present